

kmer2vec: A unified method to design baits and primers for broad-spectrum pathogen enrichment by representation learnings

Yuanyuan Wang¹, Wuming Gong², Justin Lee¹, Jan Pohl¹

¹ National Center for Emerging and Zoonotic Infectious Diseases, Centers for Disease Control and Prevention, Atlanta, GA 30329

² Department of Medicine, University of Minnesota, Minneapolis, MN 55455

Background

The metagenomic next-generation sequencing (mNGS) allows broad-spectrum pathogen detection. To ensure high-sensitivity identification for low-titre samples, two enrichment strategies are used to increase the proportion of target pathogen reads: (1) the capture probe enrichment that employs biotinylated cDNA baits molecules; (2) the PCR enrichment with primers tiled across genomes. Both require designing a minimal set of baits or PCR primers that cover majority of the target sequences.

Data

The performance of kmer2vec was evaluated on 11 pathogens through comparisons with the MSSPE method*.

Target		Number of sequences used
ZIKV	Zika virus	96
CHIKV	Chikungunya virus	479
WNV	West Nile virus	1455
EBOV	Ebola virus	1599
CCFV	Crimean-congo hemorrhagic fever virus	314
YFV	Yellow fever virus	71
LASV	Lassa virus	433
RVF	Rift valley fever virus	366
MEAV	Measles virus	126
HEV	Hepatitis E virus	303
HIV	Human Immunodeficiency Virus	500

Table. 1 Sequences downloaded from NCBI for each target pathogen

Methods

We developed a method *kmer2vec*, to learn the representations of k-mer sequences, i.e., candidate baits or primers. First, input sequences were split into overlapping segments, where the k-mers were generated. Then, we augmented the input sequences by randomly mutating specific nucleotides and encouraged the model to recover the unmutated inputs. The representations of candidate baits or primers was used to build a k-nearest neighbor (kNN) graph, followed by Louvain clustering. The decoder was used to recover the sequence of cluster centroids from the representations of their neighbors. A greedy search was used to compile a final set of primers/baits.

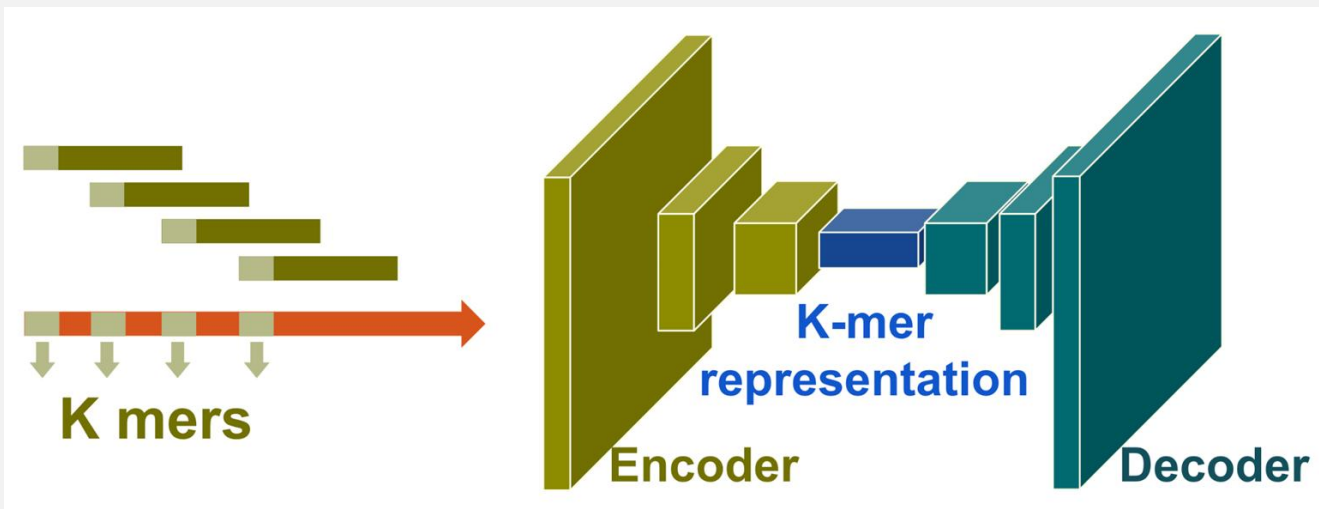


Fig. 1 Schematic representation of kmer2vec.

Results

1. On designing baits for capture probe enrichment, kmer2vec identified 1,290 120-nt baits (tolerated up to 40 nt mismatches) that cover 96.8% of the ~8,000 antimicrobial resistance genes in the MEGARes database, compared with 2,604 and 2,635 baits designed by SYOTTI** and CATCH*** respectively, representing a 50% reduction of required baits.

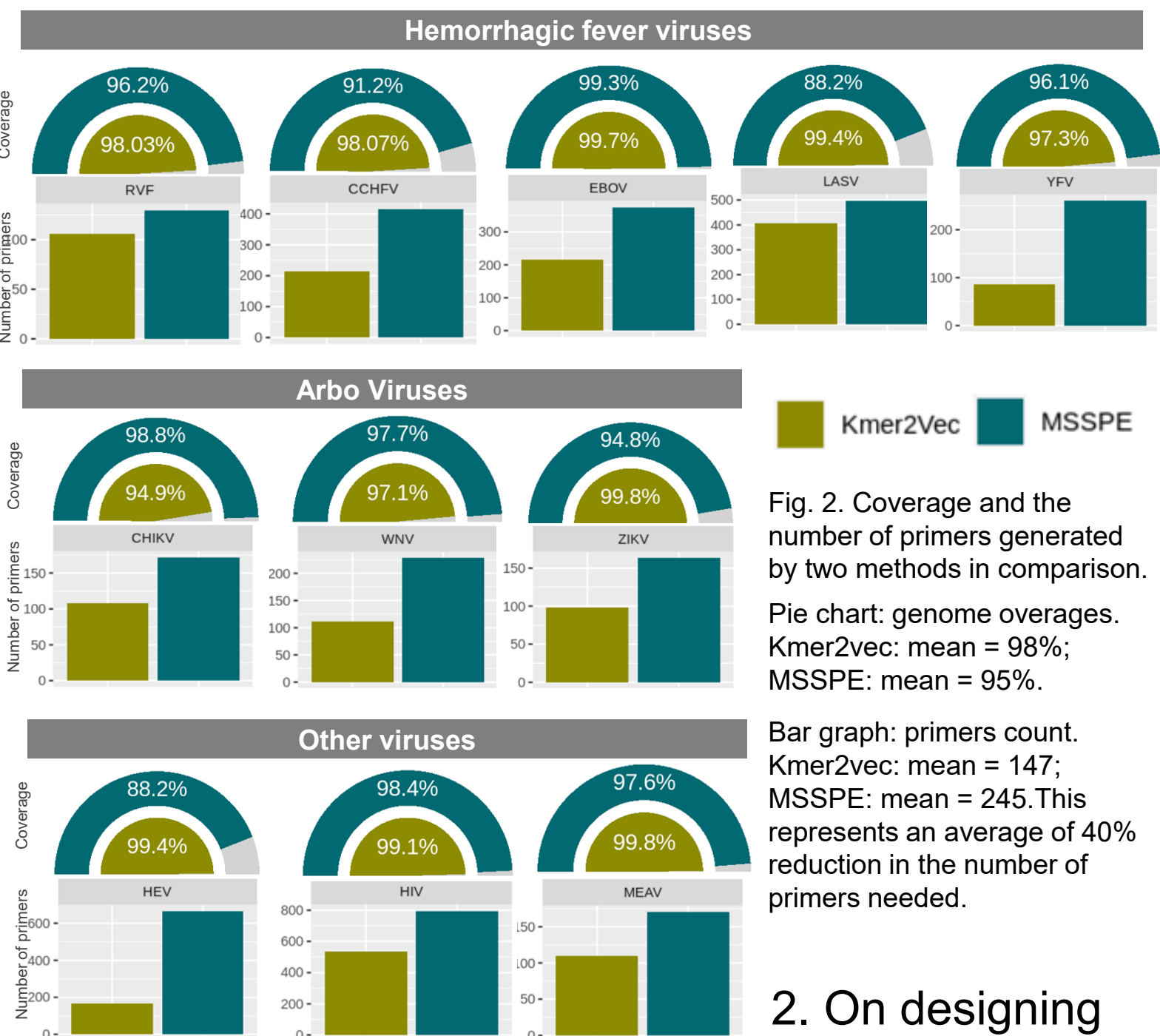


Fig. 2. Coverage and the number of primers generated by two methods in comparison. Pie chart: genome overages. Kmer2vec: mean = 98%; MSSPE: mean = 95%. Bar graph: primers count. Kmer2vec: mean = 147; MSSPE: mean = 245. This represents an average of 40% reduction in the number of primers needed.

2. On designing primer sets for PCR enrichment, kmer2vec consistently achieved high genome coverages with significantly less primers (p-value < .05). Particularly, for HIV, kmer2vec demonstrated fast learning ability by using only 500 sequences while MSSPE* used > 3000 sequences.

Conclusions

We anticipate that kmer2vec will be a valuable tool for reducing the cost and improving the efficiency of viral detection in targeted mNGS pathogen surveillance.



*Deng, et al. "Metagenomic sequencing with spiked primer enrichment for viral diagnostics and genomic surveillance." Nature microbiology 5.3 (2020): 443-454.

**Alanko, et al. "Syotti: Scalable Bait Design for DNA Enrichment." bioRxiv (2021)

***Metsky, et al. "Capturing sequence diversity in metagenomes with comprehensive and scalable probe design." Nature biotechnology 37.2 (2019): 160-168..